Multivariate Calculus Lecture Notes
Section 14.3 – Partial Derivatives

I. The idea of a partial derivative
   a. For a function of two variables, what is the value of the derivative at a particular point?
      i. In other words, what is the rate of change of the function?
      ii. Show this with contour diagram (maybe with $z = y^2 - x$?)
      iii. It depends on which direction you go.
      iv. In fact, there are infinitely many derivatives because there are infinitely many directions.
      v. We will do derivatives in general directions later; for now we focus on two specific derivatives.
   b. The *partial derivative* of $f(x, y)$ with respect to x at the point (a, b) is the rate of change of f in the direction of increasing x.
      i. Graphically, it gives the slope of the surface of f (or the slope of the line tangent to the graph of f) at the point (a, b) in the direction of increasing x.
         • Show with basketball and folder.
      ii. Notation
         1. $f_x(a, b)$
         2. $\dfrac{\partial f(a, b)}{\partial x}$
         3. $z_x(a, b)$, for $z = f(x, y)$
         4. $\left.\dfrac{\partial z}{\partial x}\right|_{(a,b)}$
      iii. Mathematically, we have

$$f_x(a, b) = \lim_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h}.$$

         1. The change is only in the x direction.
         2. This implies that y is constant when we take a partial derivative with respect to x.
   c. Similarly, there is a partial derivative with respect to y:

$$f_y(a, b) = \lim_{h \to 0} \frac{f(a, b + h) - f(a, b)}{h}.$$

II. Finding and using partial derivatives
   a. Basketball: Ask students for signs of partial derivatives at various points on the basketball.
   b. Let $f(x, y) = 4 - x^2 - 2y^2$.
      i. Graphically, at (1,1):

      ii. Find $f_x(1, 1)$.

1. We're finding the slope of the line tangent to the graph of $f(x,y) = 4 - x^2 - 2y^2$ in the direction of increasing x when $(x,y) = (1,1)$.
2. In this direction y is always 1.
3. So the equation for the parabola that I drew earlier as a function of x is actually $f(x,1) = 4 - x^2 - 2 = 2 - x^2$; it reduces to a single-variable function.
4. We have $f_x(x,1) = -2x$, so $f_x(1,1) = -2$.
5. So the slope of the surface at $(1,1)$ in the direction of increasing x is -2.

iii. Find $f_y(1,1)$.
1. Let's use the definition here.
2. We have

$$f_y(1,1) = \lim_{h\to 0} \frac{f(1,1+h) - f(1,1)}{h} = \lim_{h\to 0} \frac{4 - 1 - 2(1+h)^2 - (4 - 1 - 2)}{h}$$
$$= \lim_{h\to 0} \frac{-4h - 2h^2}{h} = \lim_{h\to 0}(-4 - 2h) = -4.$$

iv. What about a general formula that gives the partial derivatives with respect to x and with respect to y for any point on the surface?
1. In other words, what about $f_x(x,y)$ and $f_y(x,y)$ for arbitrary x and y?
2. You can use the definition, but it's easier to apply the differentiation rules you learned in first-semester calculus.
3. However, to find $f_x(x,y)$ you must treat y as constant, and to find $f_y(x,y)$ you must treat x as constant.
4. Thus $f_x(x,y) = -2x$, and $f_y(x,y) = -4y$.
5. This also yields $f_x(1,1) = -2$ and $f_y(1,1) = -4$.

c. Let $f(x,y) = \sin\left(\dfrac{x}{1+y}\right)$. Calculate $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$.

$$\frac{\partial f(x,y)}{\partial x} = \cos\left(\frac{x}{1+y}\right)\frac{1}{1+y} = \frac{1}{1+y}\cos\left(\frac{x}{1+y}\right)$$
$$\frac{\partial f(x,y)}{\partial y} = \cos\left(\frac{x}{1+y}\right)\frac{-x}{(1+y)^2} = \frac{-x}{(1+y)^2}\cos\left(\frac{x}{1+y}\right)$$

III. Using partial derivatives to approximate change

a. From the definition, $f_x(a,b) \approx \dfrac{f(a+\Delta x, b) - f(a,b)}{\Delta x}$, which is more accurate the smaller Δx is.

b. Alternatively, $f(a+\Delta x, b) - f(a,b) \approx f_x(a,b)\Delta x$.

c. This means that we can approximate a partial derivative with an average rate of change, and we can approximate changes using partial derivatives (as in first-semester calculus).

d. (#50) Use the linear approximation to estimate the percentage change in the volume of a right-circular cone of radius $r = 40$ cm if the height is increased from 40 to 41 cm.

i. Using our linear approximation formula, the change is approximately $V(40,41) - V(40,40) \approx V_h(40,40)\Delta h$.
ii. We have $\Delta h = 1$.
iii. Also, $V = \pi r^2 h/3$, so $V_h = \pi r^2/3$.
iv. Then $V_h(40,40) = \pi(40)^2/3 = 1600\pi/3$.

v. To get the percentage change, we divide this by the volume $V(40, 40) = \pi(1600)(40)/3$.
vi. Thus the percentage change is $1/40 = 2.5\%$.

IV. Partial derivatives of functions of three or more variables
a. What if we have $f(x, y, z)$?
i. Then
$$f_x(x, y, z) = \lim_{h\to 0} \frac{f(x + h, y, z) - f(x, y, z)}{h}.$$
ii. Partials of f with respect to y and z are similar.
iii. This can be extended to functions of an arbitrary number of variables.
b. Example: $f(x, y, z) = e^{xy} \ln z$
i. $f_x(x, y, z) = ye^{xy} \ln z$.
ii. $f_y(x, y, z) = xe^{xy} \ln z$.
iii. $f_z(x, y, z) = e^{xy}/z$.

V. Higher-order partial derivatives
a. Idea behind second-order partials
i. In single-variable calculus we have second derivatives.
ii. What's the corresponding idea in multivariable calculus?
1. Given $z = f(x, y)$, there are two first derivatives: $f_x(x, y)$ and $f_y(x, y)$.
2. You can differentiate $f_x(x, y)$ either with respect to x or with respect to y.
3. The same is true for $f_y(x, y)$.
4. So there are four second partial derivatives.
iii. Notation
1. We have
$$(f_x)_x \text{ or } \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) \qquad (f_x)_y \text{ or } \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$$
$$(f_y)_x \text{ or } \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) \qquad (f_y)_y \text{ or } \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right)$$
2. Normally we drop the parentheses.
- So we have f_{xy} or $\dfrac{\partial^2 z}{\partial y \partial x}$.
- This means that the orders of the variables in the two sets of notations are different from each other.

b. Interpretation of second-order partials
i. Graphically
1. $f_{xx}(a, b)$ gives the concavity of the cross-section in the x direction at (a, b).
2. $f_{yy}(a, b)$ gives the concavity of the cross-section in the y direction at (a, b).
3. $f_{xy}(a, b)$ gives the y-direction rate of change of the x-direction slopes at (a, b). It's related to how the graph is twisting in the y direction at (a, b).
4. $f_{yx}(a, b)$ gives the x-direction rate of change of the y-direction slopes at (a, b). It's related to how the graph is twisting in the x direction at (a, b).
5. Show transparency!
ii. Ask for signs of $f_x, f_y, f_{xx}, f_{yy}, f_{xy}, f_{yx}$ for the contour diagram on the transparency.
1. f_x, f_{xy}, and f_{yx} are negative.

2. f_y, f_{xx}, and f_{yy} are positive.

c. Example of calculating second-order partials
   i. Suppose $f(x, y) = x^3 + x^2y^3 - 2y^2$.
   ii. Then $f_x(x, y) = 3x^2 + 2xy^3$, and $f_y(x, y) = 3x^2y^2 - 4y$.
   iii. The second-order partials are then
      1. $f_{xx}(x, y) = 6x + 2y^3$, $f_{xy}(x, y) = 6xy^2$.
      2. $f_{yy}(x, y) = 6x^2y - 4$, $f_{yx}(x, y) = 6xy^2$.

d. Mixed partials are equal (generally-speaking).
   i. Note that $f_{xy}(x, y) = f_{yx}(x, y)$.
   ii. Clairaut's Theorem: If f_{xy} and f_{yx} are all continuous on an open disk D then $f_{xy}(a, b) = f_{yx}(a, b)$ for all points $(a, b) \in D$.

e. Third- and higher-order partial derivatives
   i. The concept can be extended to any degree of derivative; you get lots and lots of mixed partials.
   ii. The $f_{xy}(x, y) = f_{yx}(x, y)$ property holds for higher-order partial derivatives, too.
      1. As long as you take the same set of partial derivatives the order does not matter.
      2. Thus $f_{yyz}(x, y, z) = f_{yzy}(x, y, z) = f_{zyy}(x, y, z)$.
      3. This true, again, as long as all partials up to the order in question exist and are continuous at the point in question.

Homework: Section 14.3 #1, 7, 12, 13, 18, 23, 30, 33, 40, 41, 48, 49, 57, 60, 66.